

ML101

Lecture 04: Classification

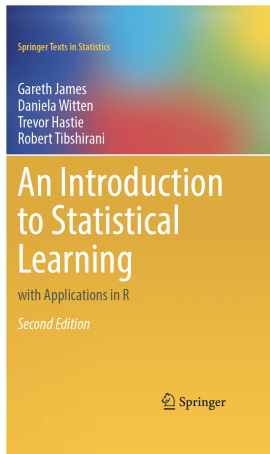
An Introduction to Statistical Learning

James, Witten, Hastie & Tibshirani

August 20, 2026

An Introduction to Statistical Learning with Applications in R

a.k.a. ISLR



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Plan

An Overview of Classification

Logistic Regression

Linear Discriminant Analysis

Classification

- Qualitative variables take values in an unordered set $\mathcal{C}$, such as:

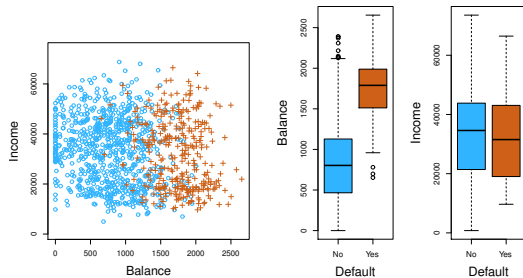
eye color $\in \{\text{brown, blue, green}\}$
email $\in \{\text{spam, ham}\}.$

- Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$, the classification task is to build a function $C(X)$ that takes as input the feature vector X and predicts its value for Y ; i.e. $C(X) \in \mathcal{C}$.
- Often we are more interested in estimating the ***probabilities*** that X belongs to each category in $\mathcal{C}$.
- For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a hard classification fraudulent or not.

Example: Credit Card Default

Default Data Set: Boxplots and scatterplots showing the relationship between balance, income, and credit card default status (No vs. Yes).

- ▶ Individuals who defaulted tended to have significantly higher credit card balances than those who did not.
- ▶ In contrast, income levels appear very similar between those who defaulted and those who did not.



Can we use Linear Regression?

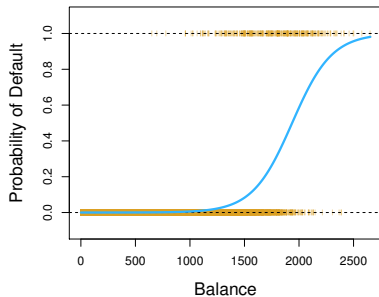
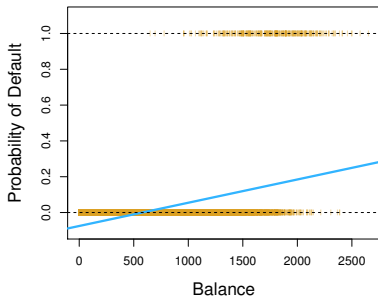
Suppose for the Default classification task that we code

$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes.} \end{cases}$$

Can we simply perform a linear regression of Y on X and classify as Yes if $\hat{Y} > 0.5$?

- ▶ In this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to linear discriminant analysis which we discuss later.
- ▶ Since in the population $E(Y \mid X = x) = \Pr(Y = 1 \mid X = x)$, we might think that regression is perfect for this task.
- ▶ However, linear regression might produce probabilities less than zero or bigger than one. Logistic regression is more appropriate.

Linear versus Logistic Regression



Linear vs. Logistic Regression Fits for Binary Data:

- ▶ **Left (Linear Regression):** A straight regression line fitted to binary 0/1 data results in predicted probabilities that dip below 0.0 for low balances and exceed 1.0 for high balances.
- ▶ **Right (Logistic Regression):** The S-shaped logistic curve naturally asymptotes at 0.0 and 1.0, guaranteeing sensible probability estimates $\Pr(Y = 1 \mid X) \in [0, 1]$ everywhere.

Linear Regression continued

Now suppose we have a response variable with three possible values. A patient presents at the emergency room, and we must classify them according to their symptoms:

$$Y = \begin{cases} 1 & \text{if stroke;} \\ 2 & \text{if drug overdose;} \\ 3 & \text{if epileptic seizure.} \end{cases}$$

- ▶ This coding suggests an ordering, and in fact implies that the difference between stroke and drug overdose is the same as between drug overdose and epileptic seizure.
- ▶ Linear regression is not appropriate here.
- ▶ Multiclass Logistic Regression or Discriminant Analysis are more appropriate.

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Logistic Regression

Let's write $p(X) = \Pr(Y = 1 \mid X)$ for short and consider using **balance** to predict **default**. Logistic regression uses the form:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

($e \approx 2.71828$ is Euler's number.)

- ▶ It is easy to see that no matter what values β_0, β_1 or X take, $p(X)$ will have values between 0 and 1.
- ▶ A bit of rearrangement gives:

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X.$$

- ▶ This monotone transformation is called the **log odds** or **logit** transformation of $p(X)$ (by log we mean natural log: $\ln$).

Maximum Likelihood

We use **maximum likelihood** to estimate the parameters:

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

- ▶ This likelihood gives the probability of the observed zeros and ones in the data. We pick β_0 and β_1 to maximize the likelihood of the observed data.
- ▶ In R we use the `glm` function with `family = binomial`.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.6513	0.3612	-29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

Making Predictions

- ▶ What is our estimated probability of default for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 1000}}{1 + e^{-10.6513 + 0.0055 \times 1000}} = 0.006$$

- ▶ With a balance of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 2000}}{1 + e^{-10.6513 + 0.0055 \times 2000}} = 0.586$$

Using `student` as the Predictor

Let's fit a logistic regression model with `student` as the single predictor:

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-3.5041	0.0707	-49.55	< 0.0001
student[Yes]	0.4049	0.1150	3.52	0.0004

Predicted probabilities:

$$\widehat{\Pr}(\text{default} = \text{Yes} \mid \text{student} = \text{Yes}) = \frac{e^{-3.5041+0.4049 \times 1}}{1 + e^{-3.5041+0.4049 \times 1}} = 0.0431,$$

$$\widehat{\Pr}(\text{default} = \text{Yes} \mid \text{student} = \text{No}) = \frac{e^{-3.5041+0.4049 \times 0}}{1 + e^{-3.5041+0.4049 \times 0}} = 0.0292.$$

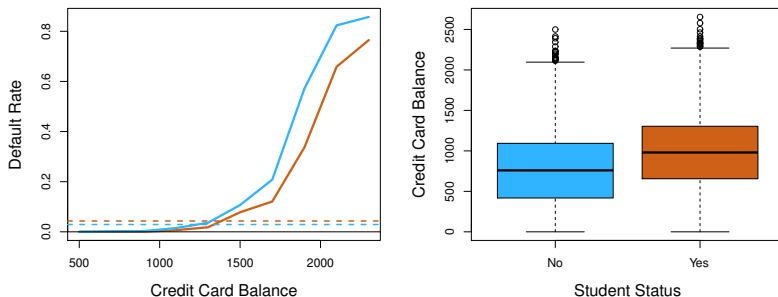
Logistic Regression with Several Variables

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \implies p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

*Why is the coefficient for **student** negative now, while it was positive in the simple model?*

Confounding in the Credit Card Data



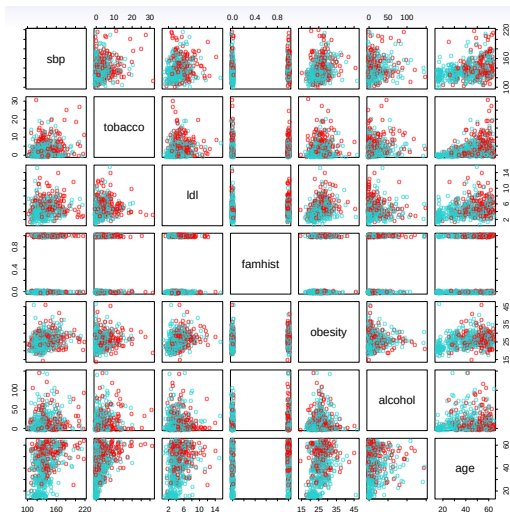
Confounding Phenomenon (Simpson's Paradox): Left (Default Rate vs. Balance): For any given credit card balance level, students (orange line) have a lower default rate than non-students (blue line). Right (Student vs. Balance Boxplot): However, students tend to hold substantially higher credit card balances on average compared to non-students.

- ▶ Students have higher overall (marginal) default rate because they carry more debt.
- ▶ But controlling for balance, being a student is associated with lower default risk.
- ▶ Multiple logistic regression teases this out!

Example: South African Heart Disease

- ▶ 160 cases of MI (myocardial infarction) and 302 controls (all male in age range 15–64), from Western Cape, South Africa in the early 1980s.
- ▶ Overall prevalence is very high in this region: 5.1%.
- ▶ Measurements on seven predictors (risk factors): **sbp** (systolic blood pressure), **tobacco**, **ldl** (low density lipoprotein cholesterol), **famhist** (family history of heart disease), **obesity**, **alcohol**, and **age**.
- ▶ Goal is to identify relative strengths and directions of risk factors.
- ▶ Part of an intervention study aimed at educating the public on healthier diets.

South African Heart Disease: Scatterplot Matrix



Scatterplot matrix of the South African Heart Disease data. The response is color coded —the cases (MI) are red, and the controls are turquoise. **famhist** is a binary variable, with 1 indicating family history of MI.

SA Heart Disease: Logistic Regression Model Output

```
> heartfit <- glm(chd ~ ., data = heart, family = binomial)
> summary(heartfit)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.1295997	0.9641558	-4.283	1.84e-05	***
sbp	0.0057607	0.0056326	1.023	0.30643	
tobacco	0.0795256	0.0262150	3.034	0.00242	**
ldl	0.1847793	0.0574115	3.219	0.00129	**
famhistPresent	0.9391855	0.2248691	4.177	2.96e-05	***
obesity	-0.0345434	0.0291053	-1.187	0.23529	
alcohol	0.0006065	0.0044550	0.136	0.89171	
age	0.0425412	0.0101749	4.181	2.90e-05	***

Null deviance: 596.11 on 461 degrees of freedom

Residual deviance: 483.17 on 454 degrees of freedom; AIC: 499.17

Case-Control Sampling and Logistic Regression

- ▶ In the SA data, there are 160 cases and 302 controls ($\tilde{\pi} = 0.35$ cases). Yet the true population prevalence is $\pi = 0.05$.
- ▶ With case-control samples, we can estimate slope parameters β_j accurately (if our model is correct); only the constant term β_0 is biased.
- ▶ We can correct the estimated intercept by:

$$\hat{\beta}_0^* = \hat{\beta}_0 + \log\left(\frac{\pi}{1-\pi}\right) - \log\left(\frac{\tilde{\pi}}{1-\tilde{\pi}}\right)$$

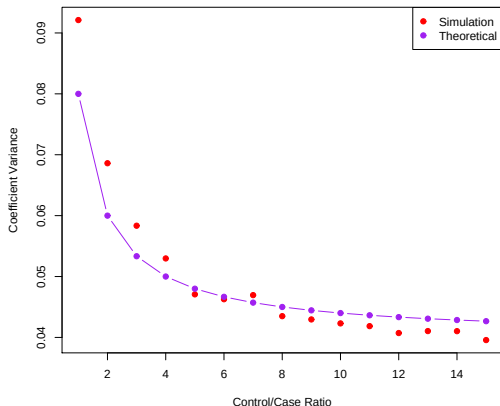
- ▶ Often cases are rare and we take all of them; up to 5 times that number of controls is sufficient.

Diminishing Returns in Unbalanced Binary Data

[Figure: Variance vs. Control/Case Ratio]

Sampling more than 5 controls per case yields diminishing returns in estimation precision.

Diminishing Returns in Unbalanced Binary Data



Sampling controls than cases reduces the variance of the parameter estimates. But after a of about 55 to 11 the variance reduction flattens out.

Control/Case Ratio vs. Coefficient Variance: As the control-to-case sampling ratio increases from 1 to 14, the variance of the parameter estimates decreases rapidly up to a ratio of about 5 : 1, after which the variance reduction flattens out.

Logistic Regression with More than Two Classes

- ▶ Generalizes naturally to $K > 2$ classes (Multinomial Logistic Regression).
- ▶ Symmetric (softmax) formulation:

$$\Pr(Y = k \mid X) = \frac{e^{\beta_{0k} + \beta_{1k}X_1 + \dots + \beta_{pk}X_p}}{\sum_{\ell=1}^K e^{\beta_{0\ell} + \beta_{1\ell}X_1 + \dots + \beta_{p\ell}X_p}}$$

- ▶ Here there is a linear function for each class.
- ▶ Redundancy: only $K - 1$ linear functions are required (one class serves as baseline).
- ▶ Multiclass logistic regression is also referred to as **multinomial regression**.

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Discriminant Analysis

- ▶ **Generative approach:** Model the distribution of X in each of the classes separately ($f_k(x)$), and then use Bayes' theorem to flip things around to obtain $\Pr(Y | X)$.
- ▶ When we use normal (Gaussian) distributions for each class:
 - ▶ Common covariance matrix $\Sigma \implies$ **Linear Discriminant Analysis (LDA)**.
 - ▶ Class-specific covariance matrix $\Sigma_k \implies$ **Quadratic Discriminant Analysis (QDA)**.
- ▶ The approach is quite general: other densities lead to other classifiers (e.g. Naive Bayes).

Bayes' Theorem for Classification

Bayes' theorem flips the conditional probabilities:

$$\Pr(Y = k \mid X = x) = \frac{\Pr(X = x \mid Y = k) \cdot \Pr(Y = k)}{\Pr(X = x)}$$

For discriminant analysis, this is written as:

$$p_k(x) = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{\ell=1}^K \pi_\ell f_\ell(x)}$$

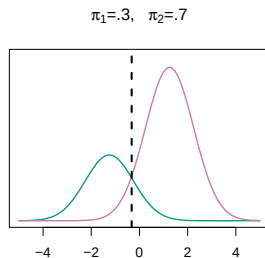
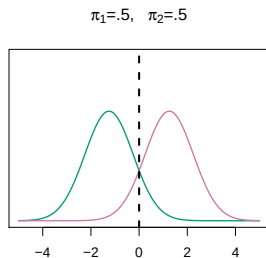
where:

- ▶ $f_k(x) = \Pr(X = x \mid Y = k)$ is the **class-conditional density** for X in class k .
- ▶ $\pi_k = \Pr(Y = k)$ is the marginal or **prior probability** for class k .

Classify to the Highest Density

1D Densities and Bayes Decision Boundary:

- ▶ **Equal Priors** ($\pi_1 = 0.5, \pi_2 = 0.5$): Decision boundary is at the point where the two density curves intersect ($x = (\mu_1 + \mu_2)/2$).
- ▶ **Unequal Priors** ($\pi_1 = 0.3, \pi_2 = 0.7$): The decision boundary shifts toward class 1 (left) to favor the higher prior probability of class 2.



Why Discriminant Analysis?

- ▶ **Separation problem:** When classes are well-separated, logistic regression parameter estimates are surprisingly unstable (coefficients diverge to $\pm\infty$). LDA does not suffer from this.
- ▶ **Small sample size:** If n is small and the distribution of predictors is approximately Gaussian in each class, LDA is more stable and efficient than logistic regression.
- ▶ **Multiclass ($K > 2$):** LDA provides convenient low-dimensional visualizations and projections of the data.

Linear Discriminant Analysis when $p = 1$

The Gaussian (normal) density is:

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2}\left(\frac{x - \mu_k}{\sigma_k}\right)^2\right)$$

Assuming shared variance $\sigma_1^2 = \dots = \sigma_K^2 = \sigma^2$, Bayes' formula gives:

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x - \mu_k}{\sigma}\right)^2\right)}{\sum_{\ell=1}^K \pi_{\ell} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x - \mu_{\ell}}{\sigma}\right)^2\right)}$$

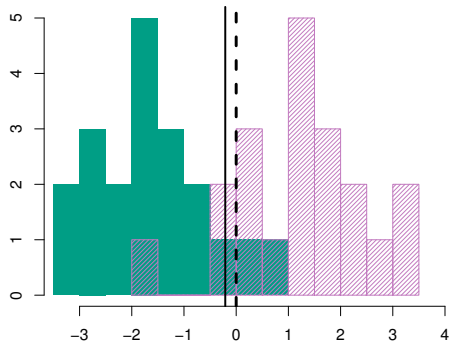
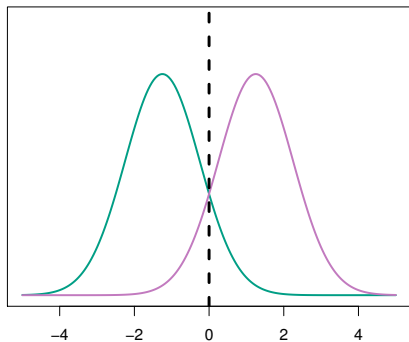
Discriminant Functions ($p = 1$)

Taking logs and removing common constants, maximizing $p_k(x)$ is equivalent to maximizing the **linear discriminant score**:

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

- ▶ $\delta_k(x)$ is a linear function of x .
- ▶ For $K = 2$ classes and equal priors ($\pi_1 = \pi_2 = 0.5$), the decision boundary ($\delta_1(x) = \delta_2(x)$) is:

$$x = \frac{\mu_1 + \mu_2}{2}$$



Example with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, and $\sigma^2 = 1$.

Typically we don't know these parameters; we just have the training data. In that case we simply estimate the parameters and plug them into the rule.

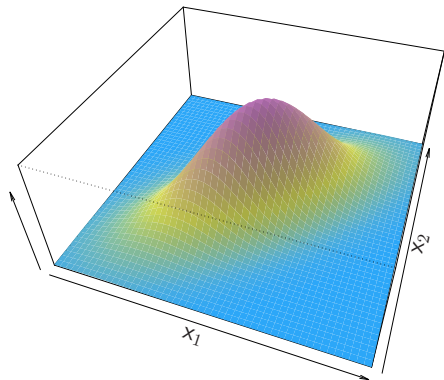
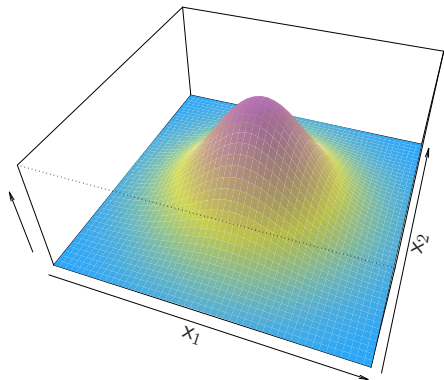
Estimating the Parameters for LDA

In practice, true parameters are unknown and estimated from training data:

$$\begin{aligned}\hat{\pi}_k &= \frac{n_k}{n} \\ \hat{\mu}_k &= \frac{1}{n_k} \sum_{i:y_i=k} x_i \\ \hat{\sigma}^2 &= \frac{1}{n-K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2 = \sum_{k=1}^K \frac{n_k - 1}{n - K} \hat{\sigma}_k^2\end{aligned}$$

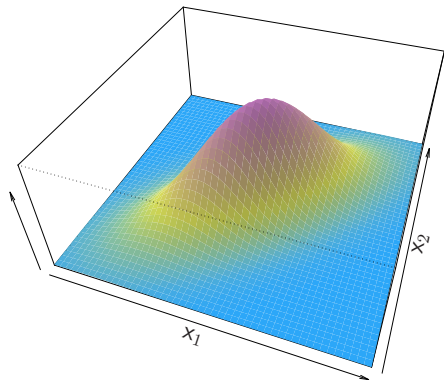
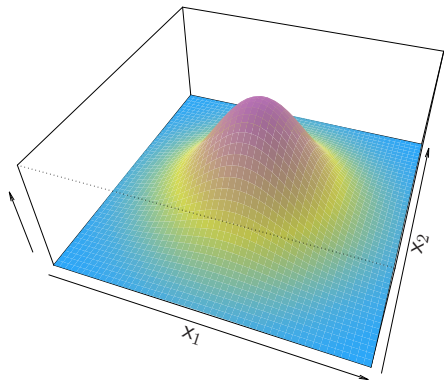
where $\hat{\sigma}_k^2 = \frac{1}{n_k-1} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$ is the sample variance in class k .

Linear Discriminant Analysis when $p > 1$



$$\text{Density: } f(x) = \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

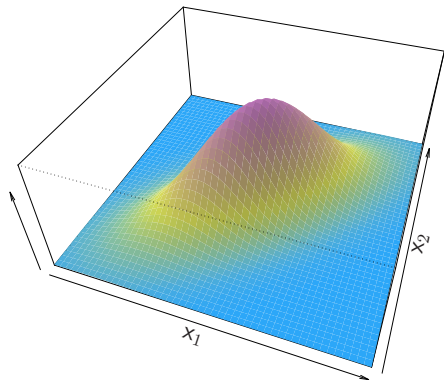
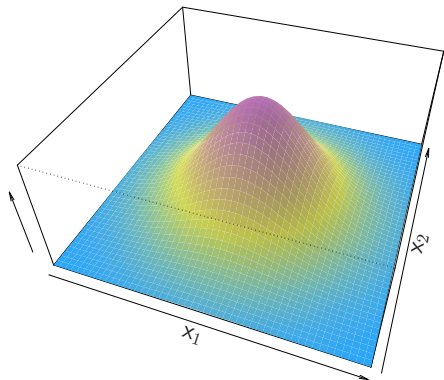
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$$\text{Discriminant function: } \delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$$

Linear Discriminant Analysis when $p > 1$



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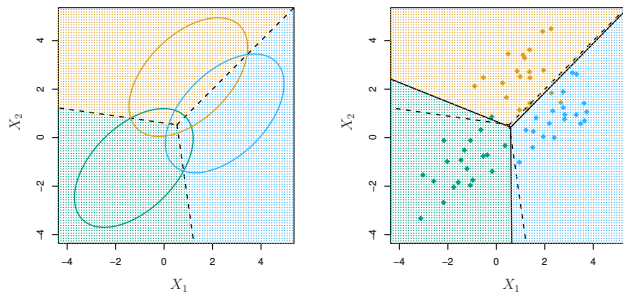
Despite its complex form,

$\delta_k(x) = c_{k0} + c_{k1}x_1 + c_{k2}x_2 + \cdots + c_{kp}x_p$ is a **linear function** of x .

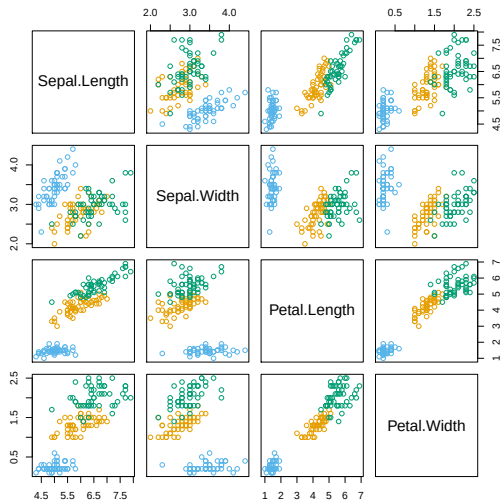
Illustration: $p = 2$ and $K = 3$ Classes

2D Gaussian LDA Contours and Decision Boundaries:

- ▶ Elliptical contour lines represent bivariate Gaussian densities for 3 classes with equal priors ($\pi_1 = \pi_2 = \pi_3 = 1/3$) and shared covariance Σ .
- ▶ Dashed lines represent optimal Bayes decision boundaries where $\delta_k(x) = \delta_\ell(x)$, dividing the plane into three polygonal regions.

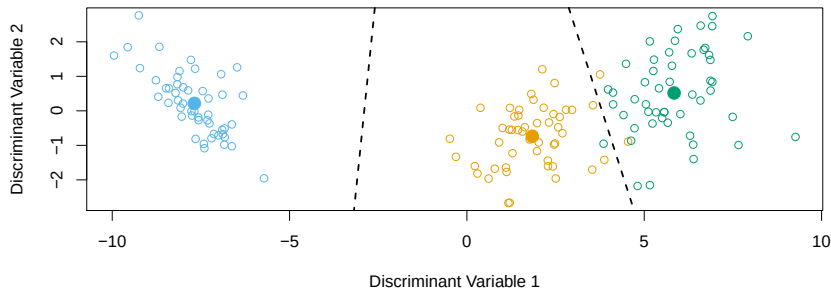


Fisher's Iris Data



- ▶ 4 continuous measurements: Sepal.Length, Sepal.Width, Petal.Length, Petal.Width.
- ▶ 3 species ($K = 3$): *Setosa* (blue), *Versicolor* (orange), *Virginica* (green) (50 samples per class, $n = 150$).
- ▶ LDA correctly classifies all but 3 of the 150 training samples (2% training error).

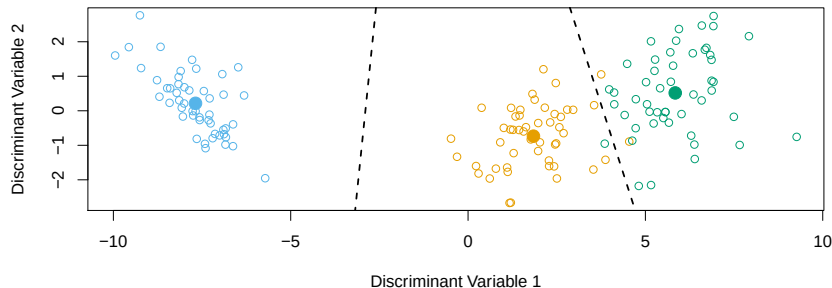
Fisher's Discriminant Plot



Fisher's Linear Discriminants for Iris Data:

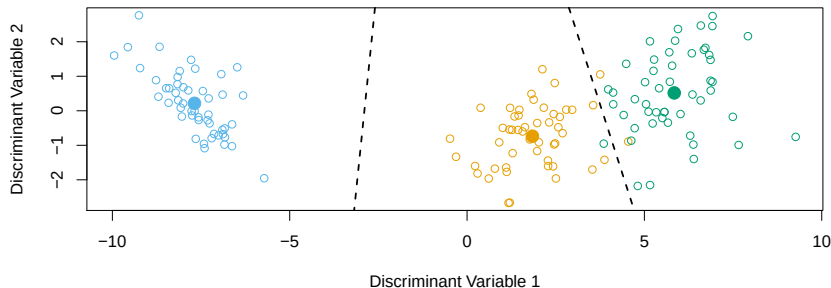
- ▶ When there are K classes, linear discriminant analysis can be viewed exactly in a $K - 1$ dimensional plot.
- ▶ Why?

Fisher's Discriminant Plot



- ▶ When there are K classes, linear discriminant analysis can be viewed exactly in a $K - 1$ dimensional plot.
- ▶ **Why?** Because LDA essentially classifies to the closest centroid in Mahalanobis distance, and K centroids span at most a $(K - 1)$ -dimensional hyperplane.

Fisher's Discriminant Plot



- ▶ When there are K classes, linear discriminant analysis can be viewed exactly in a $K - 1$ dimensional plot.
- ▶ **Why?** Because LDA essentially classifies to the closest centroid in Mahalanobis distance, and K centroids span at most a $(K - 1)$ -dimensional hyperplane.
- ▶ Even when $K > 3$, we can find the “best” 2-dimensional plane for visualizing the discriminant rule.

From $\delta_k(x)$ to Class Probabilities

Once we have estimates $\hat{\delta}_k(x)$, we convert them into posterior class probabilities:

$$\widehat{\Pr}(Y = k \mid X = x) = \frac{e^{\hat{\delta}_k(x)}}{\sum_{\ell=1}^K e^{\hat{\delta}_\ell(x)}}$$

- ▶ Classifying to the largest $\hat{\delta}_k(x)$ is identical to classifying to the class with highest posterior probability.
- ▶ For $K = 2$, we predict class 2 if $\widehat{\Pr}(Y = 2 \mid X = x) \geq 0.5$, else class 1.